

Andrea *Bonacci*

AI ENGINEER · LLM & AGENTIC SYSTEMS

I ship AI agents and RAG systems to production for companies: governed, measured, actually used.

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4+ years
of data and AI consulting for enterprise clients

40+
public repositories, 2 GitHub Marketplace Actions

Profile

From Data Scientist to **AI Engineer**, 4+ years of consulting for enterprise clients. Today I ship AI assistants to production that read documents, use company systems and ask a person for confirmation when it matters.

- **I build** governed AI agents, RAG evaluated with metrics, real-time voice AI.
- **I come from data**: pipelines, BI dashboards and NLP models, from ETL to deployment with FastAPI and Docker.
- **I teach and hire**: tutor for interns, technical interviews for Data Scientists and AI Engineers, an 8-hour university course.

Professional Experience

LINKALAB S.R.L. · CAGLIARI - part of Lynx Group since 2025 — Nov 2021 - present
Consulting company - Internal growth

AI Engineer & Data Analyst — Oct 2025 - present

- Production AI systems on client projects: real-time voice AI, agents for documents and accounting, RAG for regulatory compliance (details in Client Projects).
- Tutor for university interns: training path and final evaluation.
- Technical interviews to hire Data Scientists and AI Engineers after the acquisition.

AI Agents · RAG · Voice AI · Technical Hiring

Data Analyst & Data Scientist — Jul 2022 - Oct 2025

End-to-end data pipelines and BI dashboards (Power BI, QuickSight, Tableau, Looker Studio) for enterprise clients. NLP, sentiment analysis, time series. Deployment with FastAPI and Docker on AWS.

Python · SQL · Power BI · NLP · AWS S3/Athena

Data Scientist - internship and contract — Nov 2021 - Jun 2022

Master's thesis internship (Italian-language Emotion Analysis for online reputation), then hired with contract and first independent projects for enterprise clients.

Client Projects

On a contract basis for enterprise clients · details anonymized for confidentiality

AI Governance Platform for Agents — 2026 · in-house product

AI agents for regulated industries: identity, policy and audit on every call.

- Sandboxed agent with **19** MCP tools and **100** policy rules, fail-closed multi-user isolation; enterprise chat assistant in a client trial.
- LLM gateway with agent identity (Keycloak), policy-as-code (OPA/Rego), append-only audit trail and EU data residency.
- Multi-tenant control plane designed and built single-handedly: human-in-the-loop approvals, policy blocks, per-agent costs (FinOps).

AI Governance · OPA/Rego · LiteLLM · FastAPI · React

Differentiators

I speak business
Business degree and a Data Science master's: I start from the company's problem, not from the model.
Proof: RAG for an industrial group

AI you can control
Agents with identity, policy and audit, built for regulated industries.
Proof: AI governance platform

I teach and hire
Tutor for interns, technical hiring interviews, an 8-hour university course.
Proof: course at the University of Cagliari

Recognitions

University speaker: 8-hour course on AI for coding, University of Cagliari, April 2026.

RCE vulnerability reported: in WinBoat (open source) via Responsible Disclosure, fixed within 4 hours, credited in the v0.9.1 security release.

2 GitHub Marketplace Actions: AI PR Reviewer and AI Changelog Generator.

Skills

AI & LLM

AI Agents · MCP · Prompt Eng. · RAG · OpenAI API · Anthropic API · Tool Use · Function Calling · Document AI · Human-in-the-Loop · Realtime Voice · Sub-agents · Hugging Face · Ollama · LLM Eval · Agent Skills · DeepEval · Agentic Coding Config

GOVERNANCE & SECURITY

Policy-as-Code · OPA/Rego · Keycloak · Audit Trail · Least Privilege · Responsible Disclosure

Conversational RAG Agent for a Manufacturer's Website

Multilingual assistant for an industrial group's website, with cited sources.

- Relevant context raised from **2%** to **92%** with hybrid lexical + vector retrieval over 800+ technical documents.
- Faithfulness **1.00**, contextual recall **0.90**, no undue refusals: every answer measured with DeepEval.
- **4,000+** tests at 100% coverage; AWS Bedrock in an EU region, pgvector, deployed on EC2.

RAG · LLM Eval · AWS Bedrock · pgvector

Real-Time Conversational Voice AI Systems

Low-latency speech-to-speech AI conversation, with automated evaluation.

- OpenAI Realtime API, Gemini Live and streaming transcription (AssemblyAI), with multi-provider fallback and circuit breakers.
- RAG knowledge base queried via function calling; hallucination checks during evaluation.

Voice AI · OpenAI Realtime API · Function Calling

RAG & Document Intelligence for Compliance

Compliance of contracts and policies against a regulatory corpus, with human review.

- Hybrid search (BM25 + embeddings) and multi-stage agentic orchestration for structured verdicts with regulatory references.
- Every finding links back to its source, to block hallucinations; human-in-the-loop with checkpoints and audit trail (async FastAPI, WebSocket).

RAG · LLM Orchestration · Structured Output · Human-in-the-Loop

AI Agents for Document & Finance Automation

Automation of document and accounting processes, such as document reconciliation.

- Agents with function calling and validated output (Pydantic); extraction from XML, PDF and digitally signed files.
- Deterministic matching with tolerances and confidence scoring, ERP integration and human review of uncertain cases.

AI Agent · Document AI · Information Extraction

Personal Projects & Open Source

github.com/AndreaBonn · 40+ public repositories

Speech-to-Text github.com/AndreaBonn/speech-to-text
Semantic retrieval 100% vs 56% BM25 on 82 questions: from recorded lectures to study material with faster-whisper, a local LLM and evaluated hybrid RAG.

Claude Phone github.com/AndreaBonn/claude-phone
Claude Code driven from Telegram, with a permission gate on tools (MCP included), human approval and audit log.

AI PR Reviewer and AI Changelog Generator ai-pr-reviewer · ai-changelog-generator
2 GitHub Marketplace Actions: automated pull request review and AI-written changelogs on every release, with multi-provider LLM fallback.

Clone My Voice github.com/AndreaBonn/clone-my-voice
6 isolated TTS engines: offline voice cloning and dubbing on GPU (CUDA), Whisper, blind A/B tests.

Learn Italian github.com/AndreaBonn/impara-italiano
150 lessons from A1 to C2 with spaced repetition. One of 10 live PWAs (React, TypeScript, Firebase) with end-to-end ownership.

Also: Image Generator (SDXL and FLUX, diffusion) and video anonymization with YOLO; Claude Code Tools Explorer (TUI in Textual) and a multi-account manager for Antigravity CLI; the PWAs Meteo & Spiagge, PlotMark, BetTracker (OCR with Gemini) and apps for finance, health and shared expenses; Linux desktop tools (GTK4): remote shell via Telegram, Command Quiver, Sysbar, Nautilus extensions.

DATA SCIENCE

Python · SQL · NLP · Sentiment Analysis · Pandas · Time Series · Computer Vision

BUSINESS INTELLIGENCE

Power BI · QuickSight · Tableau · Looker Studio

MLOPS & CLOUD

Docker · FastAPI · AWS S3 · Athena · AWS Bedrock · pgvector · Flask · GitHub

Education

Master's Degree - Data Science & Business Informatics
University of Pisa, 2022
Thesis: Italian Emotion Analysis for online reputation monitoring

Bachelor's Degree - Business Administration
University of Pisa, 2018

Languages

Italian	Native
English	Fluent